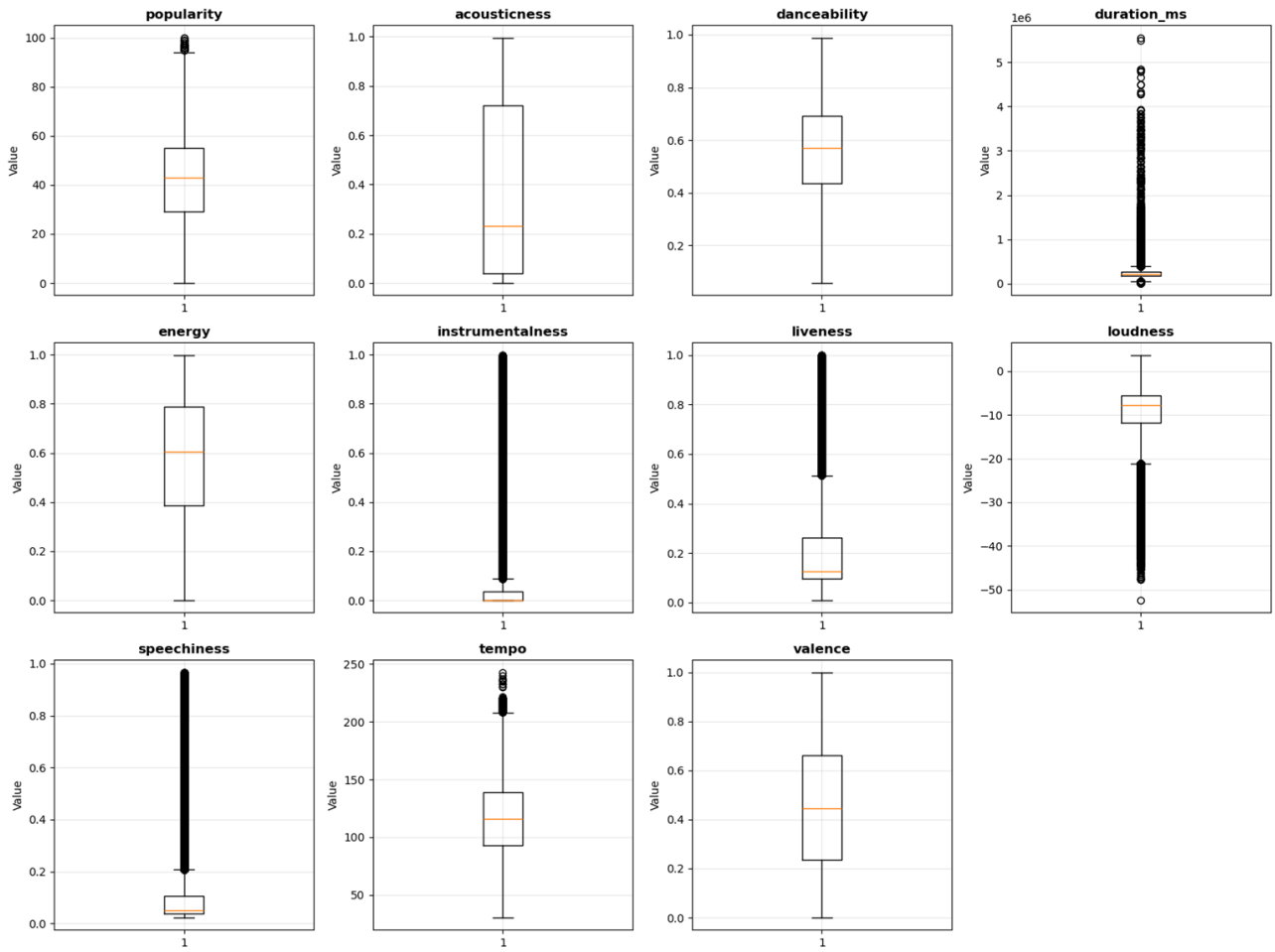
Exploratory Data Analysis (EDA)

Dataset overview

The dataset used in this project contains information on 232,725 Spotify tracks, each described through a set of audio features extracted by Spotify's internal audio analysis system. Each row in the dataset represents a single song and includes both numerical descriptors of the track's acoustic properties and categorical variables that describe some of its musical characteristics. In total, the dataset contains 18 variables. Among these, 11 are numerical features that capture different aspects of the audio signal, such as rhythm, intensity, and mood. These include variables such as danceability, which measures how suitable a track is for dancing; energy, which reflects the intensity and activity of the song; loudness, which represents the overall volume of the track; and valence, which describes the emotional positivity of the music. Other variables such as acousticness, instrumentalness, and speechiness help characterize the nature of the sound and the presence of vocals or spoken content. The dataset also includes tempo (the speed of the track), duration_ms (the track length), and the target variable popularity, which indicates how popular the track is on Spotify on a scale from 0 to 100. In addition to these numerical features, the dataset contains four categorical variables: genre, key, mode, and time_signature. These variables provide information about the musical style and structure of each track. For example, genre describes the musical category of the song, while key and mode describe its tonal structure. The time_signature indicates the rhythmic structure of the music. The dataset also includes several descriptive identifiers, such as artist_name, track_name, and track_id. Although these variables are useful for identifying songs, they were excluded from the modeling phase because they do not provide meaningful predictive information about track popularity and could potentially introduce noise into the model. An important advantage of this dataset is that it does not contain missing values, which simplifies the preprocessing stage and allows the analysis to focus directly on exploring the relationships between musical features and popularity.

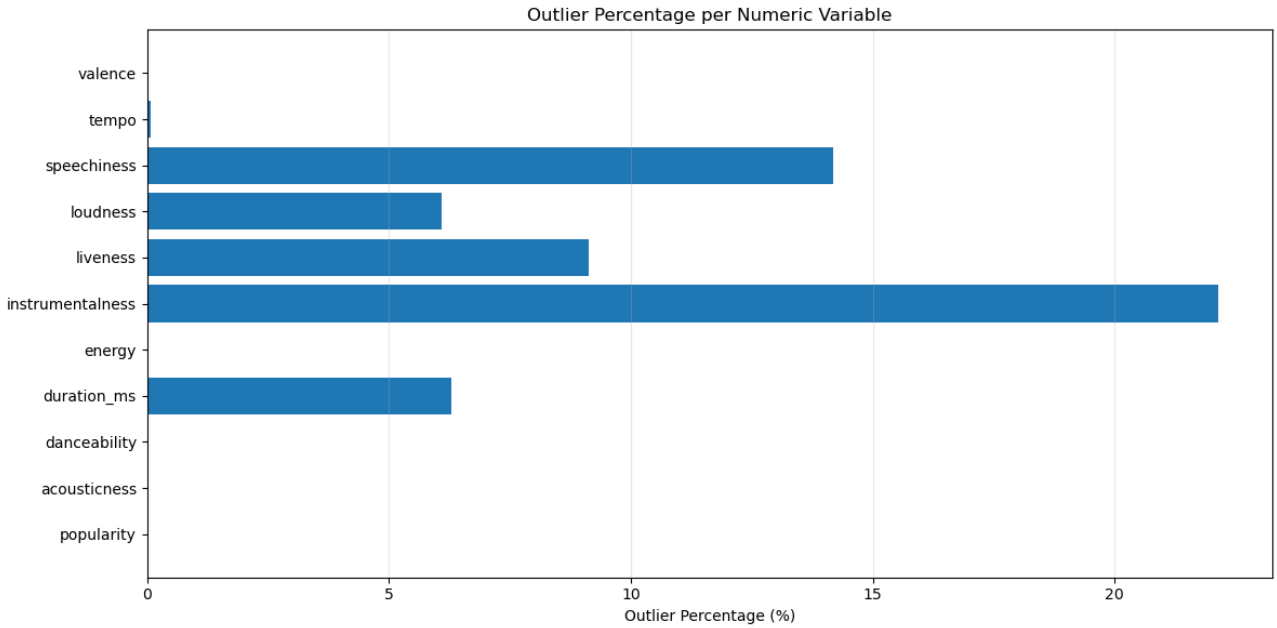
Distribution of numerical variables

To gain an initial understanding of the dataset, the numerical variables were explored using boxplots and descriptive statistics. This step helps reveal the general distribution of each variable and identify the presence of potential outliers that might influence the subsequent analysis. Most of the audio-related features in the dataset are naturally bounded between 0 and 1, as they represent normalized descriptors extracted from Spotify's audio analysis system. These variables describe different musical characteristics and tend to follow distributions that reflect common patterns in modern music production. For example, variables such as danceability, energy, and valence display relatively balanced distributions centered around intermediate values. This suggests that most tracks fall within a moderate range in terms of rhythm suitability, intensity, and emotional tone, rather than being extremely low or extremely high in these characteristics. The variable acousticness shows a broader distribution, indicating that the dataset contains a mixture of highly acoustic tracks and highly synthetic or electronically produced songs. This reflects the wide stylistic diversity present in modern music catalogs. On the other hand, speechiness and instrumentalness exhibit strong skewness toward zero. This pattern indicates that most songs contain a typical vocal structure with limited spoken-word elements, and relatively few tracks are purely instrumental or dominated by speech-like segments. A particularly interesting variable is duration_ms, which measures the length of each track in milliseconds. While the median duration is approximately 220,000 milliseconds (about 3.7 minutes) (a value consistent with the typical length of popular music tracks) the distribution shows a long right tail. The maximum duration exceeds 5.5 million milliseconds, suggesting the presence of unusually long recordings such as extended live performances, podcasts, or experimental compositions.



Outlier analysis

In order to better understand the distribution of the numerical variables and detect potential extreme values, outliers were identified using the Interquartile Range (IQR) method. This approach allows observations that fall significantly outside the typical range of the data to be flagged as potential outliers. The analysis revealed that several variables contain a relatively high proportion of outliers. In particular, instrumentalness shows the highest percentage of outliers, accounting for approximately 22.1% of the observations. This is followed by speechiness (14.2%), liveness (9.1%), duration_ms (6.3%), and loudness (6.1%). The presence of many outliers in variables such as instrumentalness and speechiness is not surprising and reflects the inherent structure of music data. Most songs in modern music catalogs contain vocals and only limited spoken content, which leads to a strong concentration of values close to zero for these features. However, certain tracks (such as instrumental compositions, live recordings, or spoken-word content) can produce much higher values, creating the observed outliers. The variable duration_ms also presents a notable number of extreme values. While most songs fall within the typical duration of a few minutes, some tracks are significantly longer, likely representing extended live performances, experimental recordings, or other atypical formats. Interestingly, the target variable popularity shows almost no outliers (approximately 0.02% of the observations). This is expected because the popularity score is bounded between 0 and 100, which naturally limits extreme values and results in a relatively stable distribution. Since the predictive models used in this project (particularly tree-based algorithms such as Random Forest) are generally robust to the presence of outliers, these observations were retained in the dataset and not removed during the preprocessing phase.



Distribution of categorical variables

In addition to the numerical audio features, the dataset includes four categorical variables that describe structural and stylistic aspects of each track: genre, key, mode, and time_signature. Analyzing the distribution of these variables helps provide a clearer picture of the musical diversity represented in the dataset.

Genre

The dataset includes 27 different musical genres, covering a wide range of styles. These range from widely popular genres such as Pop, Rock, Hip-Hop, and Electronic to more specialized or niche categories such as Opera, A Capella, and Anime. Overall, the distribution of genres appears relatively balanced across the main musical categories, although some niche genres occur less frequently. This diversity suggests that the dataset captures a broad spectrum of musical styles, which may contribute to variability in track popularity.

Musical Key

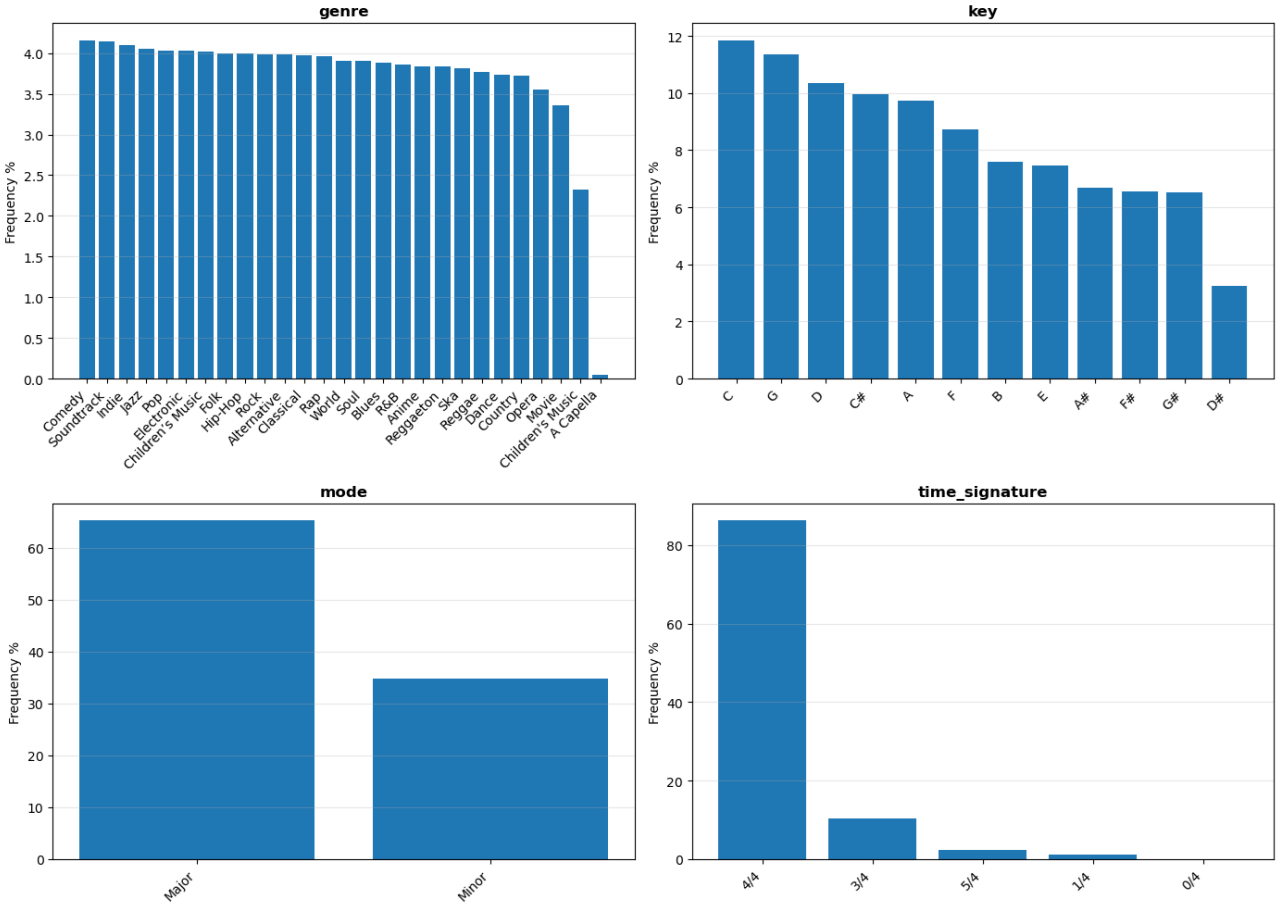
The variable key represents the tonal center of the track and includes 12 possible values, corresponding to the standard Western musical scale. The distribution of keys appears fairly uniform across the dataset, indicating that no particular key strongly dominates the collection of tracks.

Mode

The mode variable indicates whether a track is composed in a major or minor scale. In the dataset, approximately 65% of tracks are written in Major mode, while the remaining share is in Minor mode. This distribution is consistent with the characteristics of mainstream music, where major tonalities are generally more common.

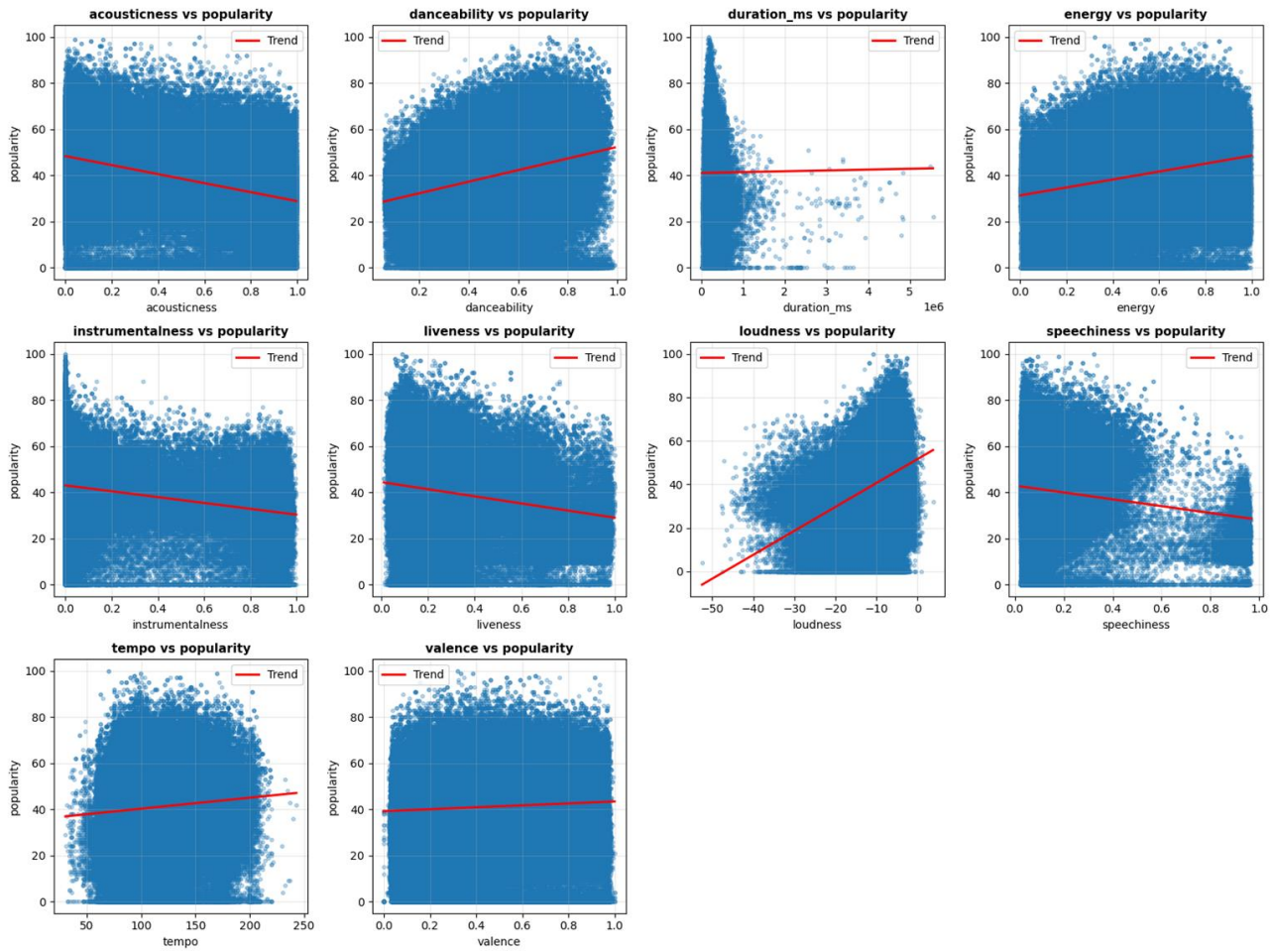
Time Signature

The time_signature variable describes the rhythmic structure of a track. As expected, the dataset is heavily dominated by the 4/4 time signature, which accounts for the vast majority of observations. This reflects a well-known feature of modern popular music, where the 4/4 rhythmic pattern is overwhelmingly prevalent. Other time signatures such as 3/4 or 5/4 appear much less frequently, representing more specialized musical compositions.



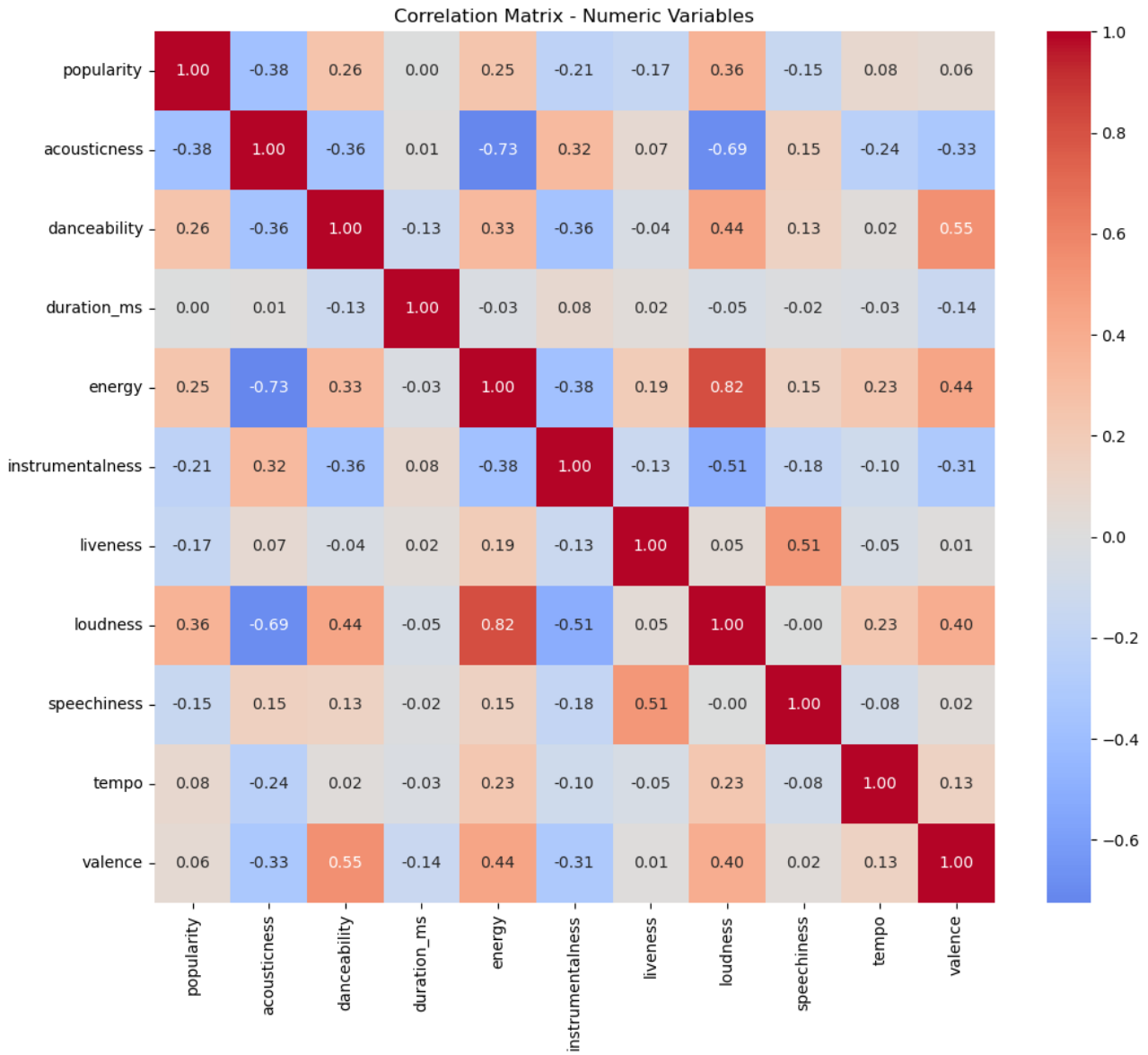
Relationship between audio features and popularity

To explore the potential relationships between the numerical audio features and the target variable popularity, several scatter plots were created. These visualizations allow us to observe how the popularity score varies as different musical characteristics change. Some interesting patterns emerge from this exploratory analysis. For instance, danceability shows a moderate positive relationship with popularity. Tracks that are easier to dance to tend, on average, to receive higher popularity scores. This pattern is consistent with the fact that many popular songs are designed to be rhythmically engaging and suitable for dance-oriented listening contexts. A similar, although slightly weaker, pattern can be observed for energy. Songs with higher energy level (often associated with louder, more intense, and more dynamic musical structures) appear to be somewhat more popular on average. Among the variables analyzed, loudness shows one of the strongest visible relationships with popularity. Tracks with higher loudness values tend to be associated with higher popularity scores. This may reflect production trends in contemporary music, where louder tracks are often perceived as more impactful or engaging. In contrast, acousticness appears to have a slight negative relationship with popularity. Tracks with very high acousticness values (typically representing acoustic or minimally produced songs) seem to be less represented among the most popular tracks in the dataset. Despite these observable trends, the scatter plots also reveal a large dispersion of points. This indicates that while some audio characteristics are associated with popularity, no single feature alone is sufficient to strongly determine how popular a track will be. Instead, popularity likely emerges from a more complex combination of musical attributes, as well as other external factors that are not captured in the dataset.



Correlation analysis

To gain a deeper understanding of the relationships between the numerical variables in the dataset, a correlation matrix was computed. This analysis helps identify how strongly different audio features are related to each other and whether any of them show meaningful associations with the target variable, popularity. Several notable relationships emerge from the correlation analysis. One of the strongest positive correlations is observed between energy and loudness (0.82). This result is intuitive, as songs that are more energetic are typically produced with higher loudness levels, reflecting the intensity and dynamics of the track. Another strong relationship appears between acousticness and energy (-0.73). This negative correlation suggests that tracks characterized by highly acoustic sounds tend to have lower energy levels, while highly energetic songs are more often electronically produced or heavily processed. A moderate positive correlation is also observed between danceability and valence (0.55). This indicates that songs with a more positive or uplifting emotional tone tend to be more danceable, which aligns with common patterns in popular music where upbeat songs often encourage movement and engagement. When focusing specifically on the relationship between the audio features and popularity, the correlations appear to be relatively weak to moderate. The strongest associations are found with loudness (0.36), followed by danceability (0.26) and energy (0.25). Although these relationships suggest that certain musical characteristics are linked to higher popularity, the relatively low correlation values indicate that no single feature alone strongly explains the popularity of a track. Instead, popularity likely depends on a combination of multiple musical attributes interacting together, as well as external factors such as marketing, artist reputation, and cultural trends that are not captured in the dataset.



Association between popularity and categorical variables

To further investigate how categorical musical characteristics relate to track popularity, ANOVA tests were conducted. This statistical method allows us to determine whether the average popularity differs significantly across the categories of each variable. The analysis reveals notable differences in the strength of association between popularity and the categorical features considered. In particular, the variable genre shows by far the strongest relationship with popularity, with an eta-squared value of 0.717. This indicates that a substantial portion of the variability in popularity can be associated with differences between musical genres. In other words, some genres tend to systematically achieve higher popularity scores than others. By contrast, the remaining categorical variables show much weaker associations with popularity. The variable time_signature presents a small effect ($\eta^2 = 0.043$), suggesting that rhythmic structure may have a limited influence on popularity. However, the effect remains relatively modest compared to the impact of genre. The variables key and mode show almost negligible associations with popularity, with eta-squared values of 0.007 and 0.005, respectively. These results suggest that tonal characteristics (such as whether a track is written in a specific musical key or in a major or minor mode) do not play a meaningful role in determining how popular a song becomes. Overall, these findings indicate that stylistic and cultural aspects of music, represented by genre, appear to influence popularity much more strongly than structural musical characteristics such as key, mode, or time signature. This aligns with the idea that listeners tend to respond primarily to

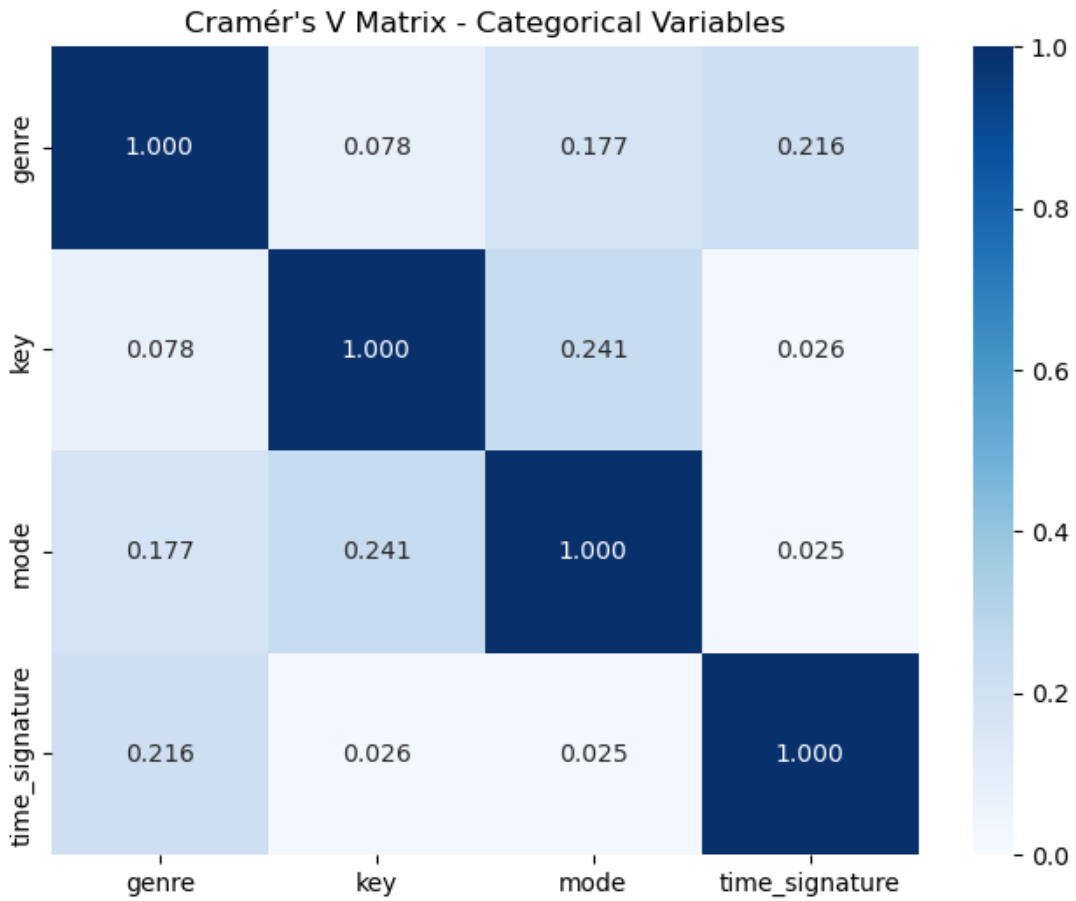
the style and context of a song rather than to technical elements of musical composition.

genre (top 10): F=26571.81, p=0.0000, eta²=0.717

key: F=153.96, p=0.0000, eta²=0.007

mode: F=1164.13, p=0.0000, eta²=0.005

time_signature: F=2624.37, p=0.0000, eta²=0.043

Association between categorical variables

To further examine how the categorical variables relate to each other, Cramér's V was calculated. This statistic is commonly used to measure the strength of association between categorical variables, with values ranging from 0 (no association) to 1 (perfect association). The results show that the associations between the categorical variables are generally weak, with most values remaining below 0.25. This indicates that the variables are largely independent from one another and do not strongly overlap in the information they capture. In practical terms, this suggests that variables such as genre, key, mode, and time signature describe different dimensions of a track's musical structure and style. For example, the genre of a song does not strongly determine its key or its rhythmic structure, and similarly, tonal characteristics such as mode do not appear to be tightly linked with the other categorical attributes. From a modeling perspective, this is a positive result. Since the categorical variables are not strongly correlated with one another, they are likely to provide complementary information rather than redundant signals. As a result, including these variables in predictive models may help capture different aspects of a song's musical identity without introducing strong multicollinearity among features.